

Development of Autonomous Humanoid Robot Control for Competitive Environment Using Fuzzy Logic and Heuristic Search

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Abstract: In this paper, we propose a novel method for interaction between humanoid robots in a competitive environment based on fuzzy logic and its implementation on an experimental platform which was validated using soccer gameplay. The proposed methodology employs vision-based feedback of the humanoid's workspace. The position of the humanoids and other objects are expressed in a newly proposed diagonal coordinate system. Hough transform with moment strategy is used to obtain odometric data of humanoids. The proposed motion planning algorithm is a combination of IDA* and linear interpolation which accounts to a major part of the path estimator. Post the path estimator, the motion execution is achieved by a fuzzy-based control system.

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1. INTRODUCTION

In recent years, the advancement in the field of automated machines has been evident in numerous industrial and research applications. Humanoid robotics has attracted the special interest of intellects all over the world. To study the dynamics of a humanoid robot and implementation of various linear, nonlinear and data driven control strategies has remained at the epicenter of the research. Dynamics of inverted pendulum problem has been considered a solved problem and effective methods have been developed to tackle both internal [Luis Sentis (2010)] and unknown foreign forces and maintain stable dynamics [Sang-Ho Hyon and Cheng (2007)]. Having considered dynamics and control of humanoid with the benchmark inverted pendulum problem, researchers and hobbyists are now putting efforts in applying various methodologies of previous studies in problems involving more competitiveness, challenge, and commercial value.

The Robo World Cup [H. Kitano and Osawa (1997); Hans-Dieter Burkhard (2002)] is a worldwide body whose aim is "To beat human soccer champions by the year 2050". Soccer is a game not only involving stable body motion, but also many other moves and runtime decision making which is considered crucial for decent game-play. A lot of research work has been done in the field of humanoid robotics regarding variety of applications also keeping sports in mind University and Branch (2013); Kos and Babi (2010). Design of the interaction between human-human and human-robot interaction can be studied to design a robot-robot game scenario. Coming after model

design and motion execution of humanoids is the data acquisition problem using computer vision and then the control design [SHI Lei (2010)]. Decision-making algorithm using machine learning and fuzzy logic based approach is one of the research subjects [Pouyan Nasrollahi (2013)]. Fuzzy logic has been studied to be one of the effective control designs which is smooth as supposed to crisp decision and even PID controllers [Albertos and Antonio (1998)]. Hybrid fuzzy logic controllers evolving from genetic algorithms for optimal design have also been worked out Sun and Er (2004). Path planning based on particle swarm optimization University and Branch (2013) has been one of the most crucial methods for the implementation. One of the best-known heuristic based methods to accessing minimal nodes is A* algorithm [Junfeng Yao and Xie (2010)]. During memory limited situations the iterative deepening A*(IDA*) is preferred which gives identical results but with little penalty on time. IDA* is a preferred solution for shortest path plan during football gaming, especially in memory bound structure. Performing inverse kinematics for motion planning has been extensively studied and many new optimized solutions have been proposed [Layale Saab and Fourquet (2013); Franck Plestan and Abba (2003)].

The paper is organized as follows: Section 2 presents the problem formulation and setup of the system. The communication model, motion constraints are explained in section 3. Section 4 comprises of the planning of the game-play of the humanoids. The whole algorithm including every aspect is covered in here briefly. The geometry of field, data

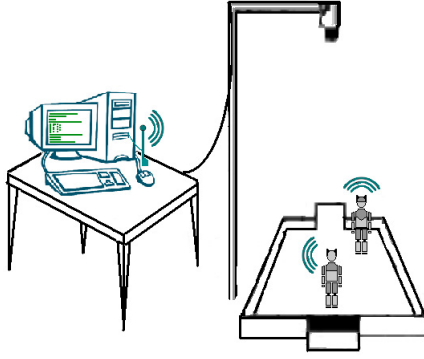


Fig. 1. Experimental Setup

acquisition and motion planning techniques are covered in sections 5, 6 and 7 respectively. The control algorithm using Fuzzy is covered in brief in section 8, which includes the input-output mapping, member function, and rule evaluation in detail. Section 9 contains the experiments and results. The future scope is covered in section 10 followed by the conclusion.

2. PROBLEM FORMULATION

The experiment consisted of 2 humanoids equipped with 2 on-head colour markers and sharp sensors (foot and mid-section), a colored ball, an off-board processor, camera and communication medium between off-board processor and humanoids. The structure of the system is inspired from previous work [Kim J (1997)] is shown in Figure 1. The computer vision system consisted of an overhead camera mounted on a truss, capturing entire field. Camera captures at a resolution of 960x720 at ~ 20 fps. The camera records data such as odometry of the robot, ball, and obstacles. The robot used for experiment was the Bioloid robot. It had 18 degrees of freedom (DOF) which can perform many complex motions as well as actions sufficient for executing a simple football game. The humanoid had an onboard micro-controller that was used to store and execute all motion steps with precision. The objective is to develop a fuzzy logic based whole-body humanoid control strategy with visual feedback for two player soccer game.

3. MOTIONS AND COMMUNICATION

Considering the bandwidth, speed optimization and taking advantage of on-board micro-controller chip all motion commands were sent in terms of action number represented in multiples of 10 with the units place representing the degree of the motion to be executed. On receiving motion code the actuators are controlled based on predefined motions programmed into the micro-controller. The motions were designed on Bioloid software Robotis platform. The decisions are then sent to respective humanoids via Zigbee protocol.

This decision is then executed by humanoid. The various motions executed by the humanoid and Zigbee values assigned to them are given in Table 1.

Each motion was given a base value. Minimum values output of each motion possible are kept as constants like *Step*, *Turn* giving values of pixels covered in one step and degrees turned in the smallest turn possible. eg: For a certain distance d ,

Motion	Zigbee sent value
Walk	10-34
Anti-clockwise turn	50-59
Clockwise turn	40-49
Small walk	9
Kick	60
Defend	61

Table 1. Motion-Zigbee command mapping

Algorithm 1 Program Flow

```

1: InitilizeCamera()
2: InitilizeZigBee()
3: InitilizeHumanoidtoHomePosition()
4: for Termination Criteria do
5:   LABEL: Repeat, GetNewFrame()
6:   DataAquasition()
7:   DecideStrikerandDefender()
8:   TrejectoryPlanning()
9:   while Ball not in Motion do
10:    GetNewFrame()
11:    DataAquasition()
12:    TrejectoryTracking()
13:    if Error Beyond Limit then
14:      Goto Repeat
15:    end if
16:    SendCommands()
17:  end while
18: end for

```

$$\text{Zigbee sent value} = 9 + \left\lfloor \frac{d}{\text{step}} \right\rfloor$$

$$\text{Full Step} = 50 \text{ pixels}, \text{ Small Step} = 20 \text{ pixels}, \\ \text{Turn} = 18^\circ$$

$$\text{Compliance for orientation} = \frac{\text{Turn}}{2} = 9^\circ$$

$$\text{Compliance for Distance} = \frac{\text{Small Step}}{2} = 10 \text{ pixels}.$$

4. GAME PLAY AND EXECUTION

Humanoids played two different roles of striker and defender according to the position of the ball. The field was divided into two parts by the half line as shown in Figure 2. The humanoid nearest to the ball is assigned as the striker and the other as the defender. The algorithm for both the humanoids run on two threads after initialization, which are almost the similar except the goal points to reach and final motion of kick or defend. The game play was designed to run in an infinite loop where the ball is passed from one quadrant to another. The algorithm is explained in Algorithm 1

Constraints to systems are enlisted below,

- No movement or decisions are taken when ball is in motion
- Ball once out of field, system halts
- Humanoids when out of field system halts
- Lighting conditions have to be fixed
- Possibly complementary colours have to be used for markers
- Humanoids only kick the ball towards goal directly, doesn't forward

5. DIAGONAL CO-ORDINATE SYSTEM

The geometric system we followed was named diagonal co-ordinate system. This enabled the code to do calculations with the perspective of each robot in mind, irrespective of the side or role, even if the side of each robot is changed. As shown in Figure 2, the left half of the field has a clockwise measurement of angles and anticlockwise measurement for the second half. This makes it convenient to test the co-ordinate system with many common data acquisition functions for any sided humanoid just by using the following transformation, (derived using the camera resolution)

$$\begin{bmatrix} X_r \\ Y_r \\ 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 960 \\ 0 & -1 & 720 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_l \\ Y_l \\ 1 \end{bmatrix}$$

6. DATA ACQUISITION

Every object has an identity by virtue of a predefined colored sticker as seen in Figure 6. The captured image is first filtered in HSV color co-ordinate system to extract predefined color patches present. Centroid of each filtered blob is calculated using $\square$, see Figure 9. The orientation of humanoid is calculated via Hough transform [Golden-shluger and Zeevi (2004)] applied on the filtered image.

Hough transform was chosen for reliability and robustness as compared to many regression techniques pardoning the computation complexity that comes with it. Hough lines provided a good estimation but with certain constraints. The colored sticker was rectangular in shape to average out all the slopes given by Hough transform which led to a single line. Even though this often gave very good results, at particular angles the averaging gave erroneous results. The problem of singularity arises when the humanoid is at 0° or 180° . Averaging of peaks in Hough space gives erroneous results at boundary especially at 0° or 180° , see Figure 3.

Principle axis can be computed using moment technique supplementary to the centroid calculation using (2). However moment method is sensitive to the shape of filtered blob and any irregularity creates false readings. Hence, a weighted average technique is adopted to improve the performance of Hough transform using (3). To expand the range of reading from $[0,180]$ to $[0,360)$, a special yellow marker is attached behind every marker to provide information about quadrant of humanoid. Once all data is collected and compiled this data can be used by the processing unit for game play strategy planning.

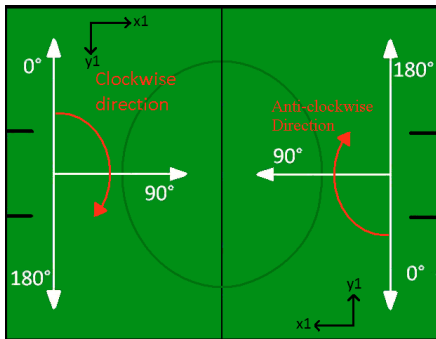


Fig. 2. Diagonal Co-ordinate System

$$m_{kj} = \sum_{(x,y) \in R} x^k y^j \quad k \geq 0, j \geq 0$$

$$x_c = \frac{m_{10}}{m_{00}} \quad y_c = \frac{m_{01}}{m_{00}} \quad (1)$$

$$\mu_{kj} = \sum_{(x,y) \in R} (x - x_c)^k (y - y_c)^j \quad k \geq 0, j \geq 0$$

$$\phi = \frac{1}{2} \text{atan2}(2\mu_{11}, \mu_{20} - \mu_{02}) \quad (2)$$

$$\text{orientation} = k_1 * \text{Hough Angle} + k_2 * \phi \quad (3)$$

$$\text{where, } k_1 = \sin^2 \phi, k_2 = \cos^2 \phi$$

7. PATH PLANNING

7.1 Shortest Path Algorithm

IDA* search seemed ideal for our situation of memory bound searching. Root node is the humanoids centroid and the goal node is destination (strike point for striker and goal post for a defender). The IDA* heuristic function [Onno Wink (2000)] was the Euclidean distance as angles which are multiples of 18° and line of sight distances between pixels which is not possible using Manhattan (will lead non-usage of diagonal paths) or Chebyshev distance (unnecessary use of diagonal motion between nodes due to equal costs of transitions). The parameters of the algorithm are,

$$\text{Heuristic} = \text{Euclidean}(\text{Node}, \text{Goal})$$

$$\text{Pathcost} = \text{Euclidean}(\text{Node}, \text{Daughter}) + \text{TurningCost}$$

$$\text{TurningCost} = 0.20 * \frac{\text{Step}}{2}$$

i.e. 20 for 50 pixel-distance jump. This is set to zero if there is no slope change since its parent. $\text{Cost} = \text{PathCost} + \text{Heuristic}$. Minimum cost route is chosen.

7.2 Neighborhood Design for Search

Algorithm parameters has to fit humanoids constrains. Minimum turn by the humanoid was limited by its construction to 18° , hence, the daughter nodes of a pixel must be 18° apart. Von Neumann neighborhood gives us 90° turns and Moore neighborhood a 45° . We needed shortest paths which were smooth curves and lines and also with a minimum 18° difference in daughters with their corresponding ancestor node. 50 pixels are crossed in one step by the humanoid and 20 pixels in a small step. So all our neighborhood pixels can be 50 pixels apart from the parent. Thus the nodes of a parent (x_p, y_p) are,

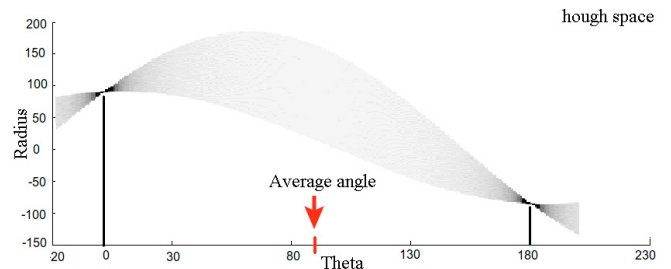


Fig. 3. Hough Accumulator Space at 0° orientation

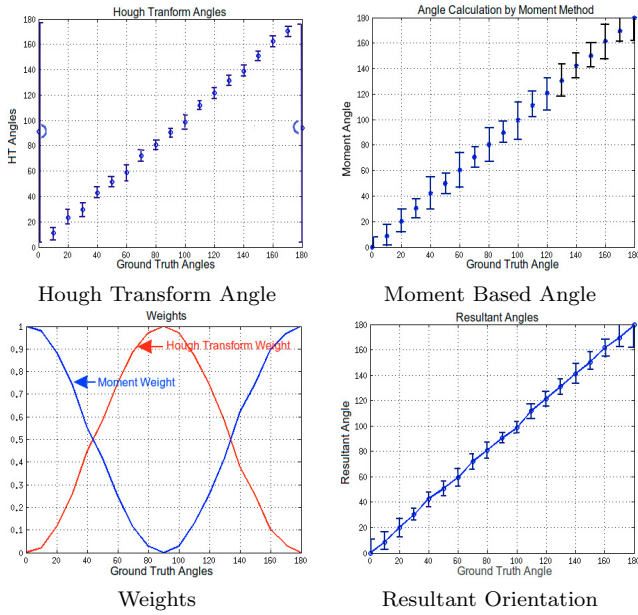


Fig. 4. Orientation computation using weighted average

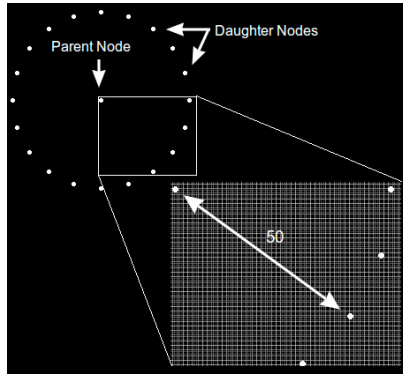


Fig. 5. IDA* Neighborhood

$$X_n = x_p + 50 * \cos(18 * n) \quad 0 \leq n < 20$$

$$Y_n = y_p + 50 * \sin(18 * n) \quad 0 \leq n < 20$$

The 20 daughter nodes for a parent pixel (center of circle) as shown in the Figure 5.

7.3 Interpolation and Redundancy removal

The pixels marked are 50 pixels apart, many of them being from the same line. The number of nodes (red markers in Figure 6) increases with the goal node being farther but sometimes unnecessarily increasing steps needed. To remove redundant nodes we interpolate the nodes by checking a change in direction. With a change detected, the parent node is marked as a transition point (green markers

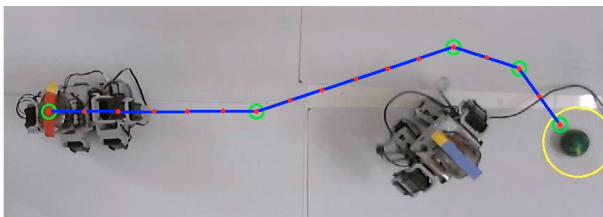


Fig. 6. Trajectory planning output on one trial

in Figure 6) and added to the queue, and if there is no change, node is removed. Hence after interpolation we have the set of nodes where the robot has to change directions. These queue of nodes are passed to the fuzzy logic code where the equations of lines between nodes and current bot position and orientation are used as input which will have an effect on the command to the humanoid.

Experimental results: We ran algorithm run separately on a case where the bot has to walk through the whole arena tackling the other bot given in Figure 6 (the path shown with the green points in Figure 6 as fixed nodes to check upon). The IDA* search and interpolation totally took 0.26ms to calculate the path.

8. MOTION PLANNING AND EXECUTION

Next step is to track the trajectory. In every tracking problem, open loop systems are unreliable and in the case of system drifts, error accumulates and produces incorrect end results. The motions of humanoid used were imprecise due to errors in actuator positions and battery level, slip w.r.t ground. Hence, it is mandatory to opt for feedback control methods for timely error correcting before leading to divergence. The feedback loop consists of the vision system, interpolated trajectory is reference input to fuzzy controller and communication system. Work has already been done in a closely related field in ball tracking using fuzzy [SHI Lei (2010)]. Fuzzy logic was primarily chosen over traditional crisp controller for following reasons,

- Data acquired from vision system is corrupted due to noise and might lead to contradicting decisions from crisp logic.
- Fuzzy logic gives same intuitive nature with a more convenient interface.
- Flexible, since only rule base change is needed for any modification in future.

8.1 Structure of Fuzzy Controller

The fuzzy controller used in this work is based on Mamdani Model of fuzzy logic controller [Mamdani (1974)]. Mamdani model does not identify system model explicitly, the controller is designed directly based on fuzzy rule base [Laxmidhar Behera (2009)]. *Input* - The fuzzy controller has 2 inputs, deviation in orientation and offset distance from planned trajectory. Since these are crisp input and fuzzy system demand fuzzy set, the crisp inputs are fuzzified so that the fuzzy controller can interpret them. *Output* - There can be three possible motion decisions taken on possible errors,

- (1) Position correction, in the case of position error. Irrespective of angle error (as angle correction will be natively included in it).
- (2) Orientation correction, in the case of orientation error only.
- (3) Re-initialize the path finding process, in case the error crossing maximum predefined bound.

8.2 Member Function

Member function for both input and output sets is defined by a trapezoidal curve for the ease calculation during de-fuzzification. The membership function plot for offset distance is given in Fig.7.

D	A	NA4		NA3		NA2		NA1		ZA		PA1		PA2		PA3		PA4	
ND2	R	-		R	-	R	-	R	-	R	-	R	-	R	-	R	-	R	-
ND1	NC	-		NC	-	NC	-	NC	-	NC	-	NC	-	NC	-	NC	-	NC	-
ZD	-	A(14)	-	A(13)	-	A(12)	-	A(11)	-	-	-	A(1)	-	A(2)	-	A(3)	-	A(4)	-
PD1	PC	-		PC	-	PC	-	PC	-	PC	-	PC	-	PC	-	PC	-	PC	-
PD2	R	-		R	-	R	-	R	-	R	-	R	-	R	-	R	-	R	-

Table 2. Fuzzy Logic Rule Base. NA = Negative angle error, PA = positive angle error, ND = Negative distance error, PD = Positive Distance Error, NC = Negative Distance Correction, R = Reset, PC = Positive Distance Correction, AC(.) = Angle Correction(Command Offset), Fuzzy membership of NAs, PAs, NDs, PDs is illustrated in Figure 7

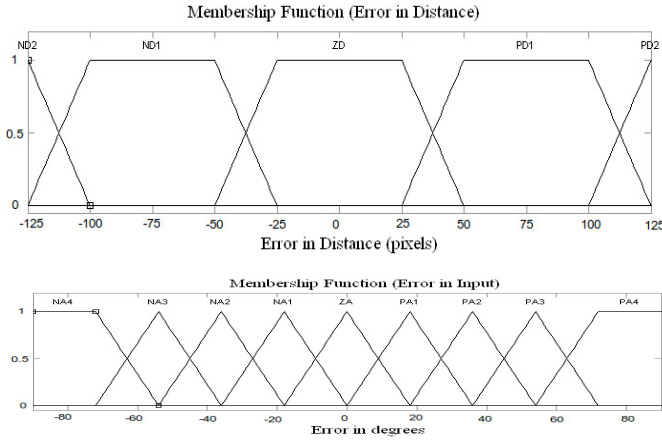


Fig. 7. Input Membership Function

The rule-base was decided based on logical motions to be performed when a particular input case is encountered. The rule-base used for the experiment is summarized in Table II.

8.3 De-fuzzification Method

Centroid method was adopted for defuzzification and gaining crisp output from the controller for further evaluation. Centroid was computed using sampled version of the formula

$$Centroid = \frac{\int x g(x) dx}{\int g(x) dx}$$

Defuzzification gave a real number output from controller for further evaluation. The number is then converted to integer value and directly added to the base value of the command and sent to the communication module. output is 2 which is added to the base value of 40 and 42 command is sent to the humanoid which corresponds to 2 turns of 18° each which gives overshoot of 6° which is acceptable. Once one node is successfully reached, other fuzzy system is given inputs w.r.t next interpolated node rather than re-initializing whole system.

9. RESULTS

The Experiment was performed on two humanoids without human interference. Images in Figure 9 show the progress of one shoot and save part of the game play recorded by the vision system. Figure 9(b) shows proper calculations of orientation and position of both humanoids. Figure 9(c) depicts the path planning using IDA*. Fuzzy logic based controller is seen to be efficiently working but with some amount of chattering in humanoid turning. Image Figure 9(d, e, f) finally shows working of controller and loop

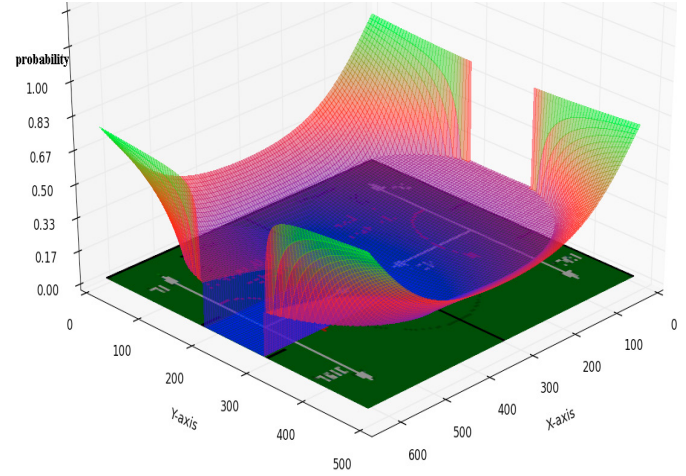


Fig. 8. Maximum Error Probability plot

termination. The images illustrate the effectiveness of designed algorithm and execution where Red humanoid effectively executes as a defender and Blue humanoid kicks the ball in precise direction after execution of all necessary motions. The game-play was found to be smooth enough to have a real time game feeling.

Over all performance of the algorithm was measured over 20 runs containing single and double kicks. The performance of IDA* in trajectory planning was found 95%, the only failure observed in longer paths due to the other bot in the way during calculation. Fuzzy Logic based controller showed 65% of successes and the failures recorded were due to chattering during turning and resetting of the algorithm due to divergence (occurrence of chattering was considered as a failed attempt). On analysis, chattering problem was seen to occur due to lack of precision in the vision system. With stable lighting conditions, the worst case probability of the ball going into the goal can be seen in fig.8)

Videos of test run whose snapshots are presented can be found online at Samant (2014b,a).

10. FUTURE SCOPE

The approach used in this work was focused on single agent and no group play technique for multi-agent [Jong-Hwan Kim and (2012)] framework was implemented. Application of Bidirectional A* search to reduce the number of nodes accessed during path planning. Intelligent controllers based on neural networks can be implemented on processing unit to compensate for non-linearity and randomness in the system.

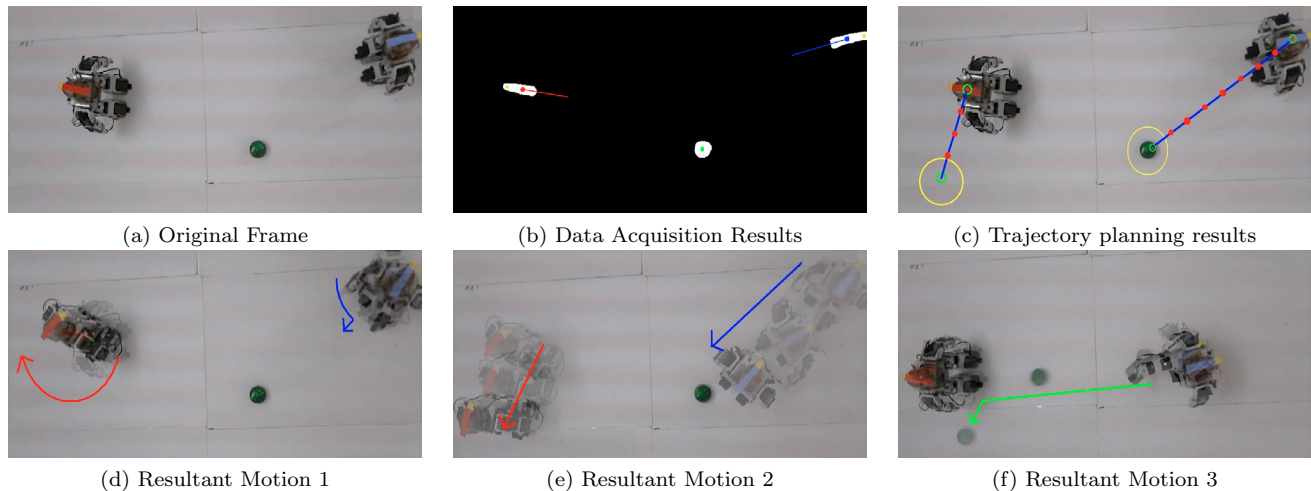


Fig. 9. Taped video images of one successful trial of kick and defend.

11. CONCLUSION

Humanoid soccer has become wide spread attraction for robotics community in recent years and will continue for years to come. In this paper, we proposed an approach based on artificial intelligence and fuzzy controller based system. The design and implementation aspects were presented briefly on whole body control of humanoids in a competitive game playing environment. Experimental results validate the proposed control strategy. The system is modular and can be easily expanded to a multi-agent system.

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